

CSCN8010: Final Workshop Submission

Multi Layer Perceptrons Workshop

Group 2 Project Contributors

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Note: Roshan did not attend the lab session and provided zero contribution to this project.

Project GitHub Repository Link

https://github.com/alicih4n/MultiLayerPerceptrons_Workshop

Executive Summary (What We Built)

This submission marks the end-to-end completion of the Multi Layer Perceptrons Workshop, heavily focusing on neural network mathematical architectures and modern frameworks.

1. Deep Learning Framework Analysis: We conducted a detailed, comparative study of Keras, PyTorch, and TensorFlow. By applying these high-level APIs to standardized image classification datasets (such as MNIST), we evaluated performance based on epoch execution time, generalization accuracy, and setup usability. Our research concluded that Keras is the most straightforward and efficient for highly standard, sequential classification architectures.

2. Low-Level NumPy Implementation (XOR Problem): To solidify our comprehension of gradient descent, we completely bypassed automated Tensor frameworks and hand-crafted a 3-layer neural network using purely **NumPy** arrays. The architecture:

- **Input:** 2 features
- **Hidden 1 & 2:** 4 nodes each using the `ReLU()` activation function to avoid vanishing gradients.
- **Output:** 1 node using `Sigmoid()` to compute probabilities.

We successfully coded stochastic forward propagation, binary cross-entropy loss calculation, backward propagation via the calculus chain-rule, and the final gradient

descent parameter update. By properly scaling the randomized weight initialization ($\ast 0.5$) and setting the learning rate to 0.1 , we demonstrated a flawless convergence on the non-linear XOR dataset in 10,000 iterations!

All executed scripts, comparisons, and documentation are provided dynamically within our notebook format inside the linked central repository.